

10. What To Wear

Best Option

- Formal dress
- Clean shoes
- Simple appearance

Avoid

- Casual slippers
 - Fancy dressing
 - Strong perfume
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11. What To Bring on Test Day

- ✓ Original CNIC
- ✓ Printed copies of degrees
- ✓ 2 passport-size photos
- ✓ Pens
- ✓ Clipboard/file folder
- ✓ NOC (if government employee)

Reach at least **30–45 minutes early**.

12. Final 5-Day Study Plan

Day 1

- DBMS
- SQL
- Keys
- Normalization

Day 2

- Networking
- OSI
- TCP/IP
- DNS/DHCP

Day 3

- OOP
- Programming concepts
- Data structures

Day 4

- Cybersecurity
- OS
- Troubleshooting

Day 5

- MCQ practice
- IQ
- Revision
- **28. DHCP is used to?**
Assign IP automatically
- **29. Router works on?**
Network Layer
- **30. Switch works on?**
Data Link Layer

- **OPERATING SYSTEM**

- **31. OS stands for?**
Operating System
- **32. Example of OS?**
Windows, Linux
- **33. Process is?**
Program in execution
- **34. Thread is?**
Lightweight process
- **35. Deadlock occurs when?**
Resources are waiting indefinitely
- **36. Paging is used for?**
Memory management
- **37. Virtual memory increases?**
Available memory
- **38. Kernel is?**
Core of OS
- **39. Multitasking means?**
Running multiple tasks

- **40. File system example?**

NTFS, FAT32

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- **OOP**

- **41. OOP stands for?**

Object Oriented Programming

- **42. Encapsulation means?**

Data hiding

- **43. Inheritance means?**

Child class gets properties

- **44. Polymorphism means?**

Many forms

- **45. Overloading is?**

Same name, different parameters

- **46. Overriding is?**

Same method in child class

- **47. Class is?**

Blueprint

- **48. Object is?**

Instance of class

- **49. Abstraction means?**

Hide implementation

- **50. Constructor is used to?**

Initialize object

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- **CYBER SECURITY**

- **51. Encryption means?**

Converting data to secure form

- **52. Hashing is?**

One-way security

- **53. Firewall protects against?**

Unauthorized access

- **54. Phishing is?**

Fake email attack

- **55. Malware means?**

Malicious software

- **56. Authentication means?**

Verify identity

- **57. Authorization means?**

Grant access

- **58. VPN is used for?**

Secure connection

- **59. HTTPS is secure because of?**

SSL/TLS

- **60. SQL Injection is?**
Database attack

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• **IQ / LOGICAL**

- **61. 5,10,15,20 ?**
25
- **62. 2,4,8,16 ?**
32
- **63. Opposite of North?**
South
- **64. 25% of 200?**
50
- **65. Average of 10,20?**
15
- **66. If $A > B$ and $B > C$ then?**
 $A > C$
- **67. Odd one: Apple, Mango, Car, Banana**
Car
- **68. 1 km = ? meters**
1000
- **69. Square root of 144?**
12
- **70. 60 seconds = ?**
1 minute

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• **ENGLISH BASICS**

- **71. Synonym of Big?**
Large
- **72. Antonym of Fast?**
Slow
- **73. Correct spelling?**
Necessary
- **74. Plural of Child?**
Children
- **75. He ____ going.**
Is

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• **ADVANCED TECH**

- **76. API stands for?**
Application Programming Interface
- **77. REST is?**
Web service architecture
- **78. JSON is?**
Data format

- **79. Cloud computing example?**
AWS, Azure
- **80. SSD is faster than?**
HDD
- **81. Cache memory is?**
Fast memory
- **82. Compiler converts?**
High-level to machine code
- **83. Linux is?**
Open-source OS
- **84. Git is used for?**
Version control
- **85. HTML is?**
Markup language
- **86. CSS is used for?**
Styling
- **87. JavaScript is?**
Programming language
- **88. RAM is?**
Temporary memory
- **89. Backup is used to?**
Data recovery
- **90. BIOS is stored in?**
ROM

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• **FINAL MIX**

- **91. LAN stands for?**
Local Area Network
- **92. WAN stands for?**
Wide Area Network
- **93. Motherboard is?**
Main circuit board
- **94. USB stands for?**
Universal Serial Bus
- **95. Excel is used for?**
Spreadsheets
- **96. Word is used for?**
Documents
- **97. PowerPoint is used for?**
Presentations
- **98. Database stores?**
Organized data
- **99. Compiler error occurs during?**
Compilation
- **100. Algorithm is?**
Step-by-step procedure

- **HARD LEVEL MCQs (With Answers)**

- **DATABASE / SQL (Advanced)**

- **1. Which normal form removes transitive dependency?**

A) 1NF
B) 2NF
C) 3NF
D) BCNF

Answer: C

- **2. In DBMS, ACID property ensures?**

A) Speed
B) Reliability of transactions
C) Encryption
D) Compression

Answer: B

- **3. Which isolation level prevents dirty reads?**

A) Read Uncommitted
B) Read Committed
C) Snapshot
D) None

Answer: B

- **4. Which SQL clause is executed first logically?**

A) WHERE
B) SELECT
C) FROM
D) GROUP BY

Answer: C

- **5. What does an index mainly improve?**

A) INSERT speed
B) DELETE speed
C) SELECT speed
D) Storage

Answer: C

- **6. Which join returns unmatched records?**

A) Inner Join
B) Cross Join
C) Left Join
D) Self Join

Answer: C

- **OPERATING SYSTEM (Advanced)**

- **7. Which scheduling algorithm may cause starvation?**

A) FCFS
B) Round Robin

- C) Priority Scheduling
- D) None

Answer: C

- **8. Deadlock requires how many conditions?**

- A) 2
- B) 3
- C) 4
- D) 5

Answer: C

- **9. Paging avoids?**

- A) Fragmentation
- B) Deadlock
- C) Encryption
- D) Virtual memory

Answer: A

- **10. Which memory is fastest?**

- A) RAM
- B) Cache
- C) SSD
- D) HDD

Answer: B

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• **NETWORKING (Advanced)**

- **11. Which protocol is connectionless?**

- A) TCP
- B) FTP
- C) UDP
- D) HTTPS

Answer: C

- **12. What does subnet mask do?**

- A) Encrypt data
- B) Divide network & host
- C) Assign IP
- D) Block malware

Answer: B

- **13. Which layer handles routing?**

- A) Data Link
- B) Network
- C) Transport
- D) Session

Answer: B

- **14. ARP is used to?**

- A) Convert IP to MAC
- B) Convert MAC to IP
- C) Encrypt packets

D) Route traffic

Answer: A

- **15. Which device works at Layer 3?**

A) Switch

B) Hub

C) Router

D) Repeater

Answer: C

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• **OOP (Advanced)**

- **16. Runtime polymorphism is achieved by?**

A) Overloading

B) Overriding

C) Encapsulation

D) Abstraction

Answer: B

- **17. Which OOP concept binds data and functions?**

A) Inheritance

B) Polymorphism

C) Encapsulation

D) Aggregation

Answer: C

- **18. Abstract class can have?**

A) Only abstract methods

B) Only concrete methods

C) Both abstract & concrete

D) No methods

Answer: C

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• **CYBER SECURITY**

- **19. Hash function property?**

A) Reversible

B) One-way

C) Slow encryption

D) Symmetric

Answer: B

- **20. Which attack floods server?**

A) Phishing

B) DDoS

C) SQL Injection

D) Spoofing

Answer: B

- **21. Public key cryptography uses?**

A) One key

B) Two keys

- C) No key
- D) Password

Answer: B

- **22. SSL/TLS mainly provides?**

- A) Speed
- B) Encryption
- C) Storage
- D) Backup

Answer: B

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• **DATA STRUCTURES**

- **23. Stack follows?**

- A) FIFO
- B) LIFO
- C) Random
- D) Priority

Answer: B

- **24. Queue follows?**

- A) LIFO
- B) FIFO
- C) Tree
- D) Graph

Answer: B

- **25. Binary search time complexity?**

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n^2)$
- D) $O(1)$

Answer: B

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• **ANALYTICAL / LOGICAL (Hard)**

- **26. If all A are B, and all B are C, then?**

- A) Some A are C
- B) All A are C
- C) No relation
- D) None

Answer: B

- **27. Find next number: 3, 7, 15, 31, ?**

Pattern: $\times 2 + 1$

Answer: 63

- **28. If 20% of X = 50, X = ?**

X = 250

- **29. Ratio 2:3, total 25. First value?**

10

- **30. Probability of head in fair coin?**

1/2

-

• **VERY HARD CONCEPTUAL**

- **31. Which is NOT a database transaction property?**

- A) Atomicity
- B) Consistency
- C) Isolation
- D) Redundancy

Answer: D

- **32. Which memory is non-volatile?**

- A) RAM
- B) Cache
- C) ROM
- D) Register

Answer: C

- **33. Which protocol is used for secure file transfer?**

- A) FTP
- B) HTTP
- C) SFTP
- D) Telnet

Answer: C

- **34. Which algorithm is used in encryption like RSA based on?**

- A) Prime factorization
- B) Sorting
- C) Searching
- D) Hashing

Answer: A

- **35. Virtual memory uses?**

- A) Hard disk
- B) Cache
- C) CPU
- D) ROM

Answer: A

INTERVIEW TECHNICAL QUESTIONS

1. Database & Data Management

1. Explain normalization and why it is important.

They expect:

- Reduce redundancy
- Improve data integrity

- 1NF, 2NF, 3NF basics
-

2. What are ACID properties? Why are they important in government systems?

They expect:

- Atomicity
 - Consistency
 - Isolation
 - Durability
 - Ensures reliable transactions (very important for citizen data)
-

3. How would you design a secure database for national identity data?

Good points:

- Encryption at rest
 - Role-based access control
 - Audit logs
 - Backup & disaster recovery
 - Data masking
-

4. Difference between SQL and NoSQL databases?

They expect understanding of:

- Structured vs unstructured
 - Transactions
 - Scalability
-

2. Networking & Infrastructure

5. Explain TCP vs UDP. When would you use each?

They want:

- Reliability vs speed
 - Use cases
-

6. What is subnetting and why is it used?

7. How does DNS work?

8. What steps will you take if a government office network goes down?

Good answer structure:

1. Check physical connectivity
 2. Check router/switch
 3. Check IP configuration
 4. Ping test
 5. Check firewall
 6. Escalate if needed
-

3. Cyber Security (VERY IMPORTANT for NADRA)

9. How would you protect sensitive citizen data?

Mention:

- Encryption
 - Access control
 - Multi-factor authentication
 - Firewall
 - Regular audits
 - Intrusion detection system
-

10. What is SQL Injection? How do you prevent it?

Prevention:

- Parameterized queries
 - Input validation
 - Prepared statements
-

11. What is the difference between authentication and authorization?

12. What is a zero-day vulnerability?

 4. System Design / Practical Thinking

13. How would you design a high-availability system for national database services?

Mention:

- Load balancing
 - Redundancy
 - Failover servers
 - Cloud backup
 - Disaster recovery plan
-

14. What is backup strategy? Explain types.

- Full backup
 - Incremental backup
 - Differential backup
-

15. How would you handle a data breach incident?

Good structure:

1. Isolate system
 2. Identify source
 3. Inform management
 4. Fix vulnerability
 5. Restore from backup
 6. Audit logs
-

 5. Operating Systems

16. What is process vs thread?

17. What is deadlock? How can it be prevented?

Prevention:

- Avoid circular wait
 - Resource ordering
 - Deadlock detection
-

6. Leadership & Management (Very Important)

Since it's Deputy level:

18. How will you manage technical team under pressure?

They want:

- Communication
 - Task delegation
 - Monitoring
 - Documentation
-

19. If a junior staff makes repeated technical mistakes, what will you do?

Good answer:

- Train them
 - Guide them
 - Provide feedback
 - Maintain professionalism
-

20. How do you ensure data confidentiality in government environment?

Mention:

- Security policies
 - Access levels
 - Monitoring
 - Compliance standards
-

Scenario-Based Questions (Very Important)

These are common:

21. Server is slow — what will you check first?

- CPU usage
 - RAM
 - Disk space
 - Network traffic
 - Logs
-

22. System is not responding in multiple offices — what could be the issue?

- Central server problem
 - Network failure
 - Firewall issue
 - Power failure
-

23. How would you migrate data from old system to new system safely?

- Backup
 - Testing environment
 - Data validation
 - Gradual migration
 - Rollback plan
-

They May Also Ask:

- What is your experience with databases?
 - Have you worked with network security?
 - What programming languages do you know?
 - What is your role in previous projects?
 - Why should we hire you?
-

How To Answer Professionally

Use this structure:

- 1. Define concept briefly**
- 2. Explain importance**
- 3. Give practical example**
- 4. Relate to government system**